Lab 06 Contribution Journal: Regression and Classification

ITAI 1371 · Team TheBetaThetas
Members: Brandy Gonzales, Derrick K. Gibbs-McGlaston, Victor Djuma, Marlon Roberts, Pablo Lara

Brandy Gonzales

I completed the Lab 6 notebook on the Titanic dataset. I built a linear regression model to predict Fare, which had an RMSE of about $58, and a logistic regression model to predict Survived, which was 74.83% accurate. I fixed errors along the way, including a missing numpy import and a NaN error from running the Sex mapping cell twice. I answered all three Knowledge Check questions and wrote the reflective journal. I saved my work as a PDF and sent it to our team leader, Marlon Roberts, who submits it on our behalf.

Derrick K. Gibbs-McGlaston

This entry describes my own work on the Lab 04 notebook. The assignment is a group project, but I worked through the full notebook in my own Colab session. I am reporting only the work I can account for, rather than assigning credit to other group members.

I ran the regression section using the Titanic dataset. I checked the training and testing split and reviewed the linear regression output. The notebook showed 571 training records and 143 testing records. I reviewed the mean squared error of 3364.92, the root mean squared error of $58.01, and the model coefficients to understand what the fare predictions showed.

I also ran the classification section and reviewed its output. The logistic regression model used age, passenger class, and sex to predict survival. I checked the reported accuracy of 74.83%, which represented 107 correct predictions out of 143 test records. I compared that result with the regression evaluation to understand why the two tasks use different measures.

I reviewed the notebook knowledge-check answers on the difference between regression and classification, what the linear regression coefficients mean, and why mean squared error and accuracy are used for different problems. I also exported the executed notebook to PDF with the group name in the filename so that the code and results could be submitted together.

My contribution journal and reflection explain what I learned from running and reviewing the notebook. I am not claiming that I wrote every original line of the notebook code or that I completed work performed by other team members. If the group divides the submission into assigned sections, my recorded contribution should be matched to the sections I actually submit.

Victor Djuma

For this lab, my individual contribution was completing the notebook in Google Colab and making sure the required machine learning models were trained and evaluated correctly. I worked through both the regression and classification sections and reviewed the outputs so I could understand what each model was doing.

For the regression part, I used Age and Pclass as input features and Fare as the target. I created a Linear Regression model, trained it with the training data, made predictions using the test data, and evaluated the results with Mean Squared Error. My model produced an MSE of 3364.92. I also printed the model coefficients

and obtained approximately -0.44 for Age and -38.30 for Pclass. This helped me understand that coefficients show how each feature influences the predicted value.

For the classification part, I used Age, Pclass, and Sex as the input features and Survived as the target. I created a Logistic Regression model, trained it, made survival predictions, and evaluated the results using accuracy. The model achieved an accuracy of 74.83%, showing that it correctly predicted survival for about three out of every four passengers in the test data.

I also completed the Knowledge Check questions. I explained the difference between regression and classification, interpreted the Linear Regression coefficients, and explained why Mean Squared Error is appropriate for predicting numerical values while accuracy is appropriate for classification outcomes.

Finally, I reviewed the notebook to make sure the required code cells executed successfully, removed unused placeholder cells, and saved the completed work as both a Jupyter notebook and a PDF. My contribution focused on completing the required coding tasks, checking the results, understanding the model outputs, and preparing the finished work for submission.

Marlon Roberts

For this lab, we learned and contributed to regression and classification. We were resolving environment restrictions to get the raw data loaded and accessible for Python. Checking out the initial dimensions and structural characteristics of the dataset. Debugging the environment for faults and handling package gaps that blocked the execution flow. Plotting the data variations and compiling the ultimate file for submission guidelines. While doing all that, I submit and check the format of the work and make sure it gets submitted. We go through all the steps in the lab.

Pablo Lara

Contribution not yet received.